

Interpreting the FLS Wage Entropy Diagnostics

The diagnostics file contains five rows for each AEWR region-year:

$$\text{gap closure} \in \{0, 0.25, 0.50, 0.75, 1\}.$$

Read these rows together as a path from the original BEA weights to weights that exactly match the FLS wage.

Wage columns

fls_target_wage

The published preliminary FLS field-and-livestock wage:

$$W_{rt}^{FLS}.$$

oews_prior_weighted_wage

The OEWS wage calculated using the original BEA-area weights:

$$W_{rt}^0 = \sum_a q_{art} x_{at}.$$

The full baseline discrepancy is:

$$W_{rt}^{FLS} - W_{rt}^0.$$

A positive discrepancy requires shifting weight toward higher-wage areas. A negative discrepancy requires shifting weight toward lower-wage areas.

gap_closure

The requested fraction of the baseline discrepancy to close.

- 0: retain the original BEA weights.
- 0.25: close 25% of the discrepancy.
- 0.50: close 50%.
- 0.75: close 75%.
- 1: match the FLS wage exactly.

entropy_target_wage

The wage target for a particular row:

$$W_{rt}^\gamma = W_{rt}^0 + \gamma (W_{rt}^{FLS} - W_{rt}^0).$$

oews_calibrated_weighted_wage

The OEWS wage after entropy balancing:

$$W_{rt}^C(\gamma) = \sum_a \hat{\omega}_{art}^\gamma x_{at}.$$

For a successful calibration, this should equal **entropy_target_wage**.

oews_sampling_composition_shock

The wage change generated by reweighting:

$$\hat{\eta}_{rt}^\gamma = W_{rt}^C(\gamma) - W_{rt}^0.$$

At full closure, this equals the complete FLS–OEWS discrepancy. At intermediate closure levels, it is a shrunk version of that discrepancy.

oews_fls_wage_gap_remaining

The part of the FLS–OEWS discrepancy left unexplained after calibration:

$$W_{rt}^{FLS} - W_{rt}^C(\gamma).$$

achieved_gap_closure

The fraction of the discrepancy actually closed:

$$\frac{W_{rt}^C(\gamma) - W_{rt}^0}{W_{rt}^{FLS} - W_{rt}^0}.$$

For feasible cells, this should approximately equal **gap_closure**.

Feasibility columns

calibration_status

- **calibrated**: a finite solution was found.
- **calibrated_boundary**: the target equals the lowest or highest available OEWS wage.
- **outside_support**: the target is outside the OEWS wage range.
- **no_wage_variation**: all available OEWS areas have the same wage.

oews_minimum_wage and oews_maximum_wage

These define the wage range attainable with nonnegative weights. Targets near either boundary generally require concentrated weights.

Weight-deviation columns

entropy_lambda

The entropy weights satisfy:

$$\hat{\omega}_a^\gamma = \frac{q_a \exp(\lambda^\gamma x_a)}{\sum_b q_b \exp(\lambda^\gamma x_b)}.$$

- Positive lambda: upweight higher-wage areas.
- Negative lambda: upweight lower-wage areas.
- Zero: retain the BEA prior.

entropy_kl_divergence

The total departure from the prior weights:

$$D_{KL} = \sum_a \hat{\omega}_a \log \left(\frac{\hat{\omega}_a}{q_a} \right).$$

KL divergence should equal zero at **gap_closure** = 0 and generally increase along the path. A sharp increase near full closure means the last portion of the wage gap is expensive to match.

prior_effective_area_count

The effective number of OEWS areas under the BEA prior:

$$N^{eff,0} = \frac{1}{\sum_a q_a^2}.$$

calibrated_effective_area_count

The effective number after calibration:

$$N^{eff,\gamma} = \frac{1}{\sum_a (\hat{\omega}_a^\gamma)^2}.$$

effective_area_count_ratio

The calibrated effective count relative to the prior:

$$\frac{N^{eff,\gamma}}{N^{eff,0}}.$$

- Near 1: preserve prior concentration.
- 0.50: retain half of the prior effective count.
- Near 0: calibration concentrates weight on very few areas.

- Above 1: calibration makes weights more evenly distributed than the prior.

maximum_weight_adjustment

The largest multiplicative increase in an area's weight:

$$\max_a \frac{\hat{\omega}_a^\gamma}{q_a}.$$

A large value can occur for an area with a very small prior, so interpret it together with the maximum calibrated weight.

maximum_calibrated_area_weight

The largest final area weight:

$$\max_a \hat{\omega}_a^\gamma.$$

A value of 0.50 means that one OEWS area contributes half of the calibrated regional wage.

Coverage columns

oews_observed_prior_mass

The share of the original BEA prior associated with areas having usable OEWS wages. Values near one indicate strong coverage. Low values mean that the observed areas were substantially renormalized.

oews_area_count

The number of OEWS areas included in the calibration.

oews_distinct_wage_count

The number of distinct wage values available across those areas. This may be smaller than the area count when multiple areas report the same wage.

Reading the path

A favorable path has:

- successful calibration through full closure;
- slowly increasing KL divergence;
- an effective-area-count ratio remaining near one;
- moderate maximum weight adjustments;
- no dominant calibrated area;

- observed prior mass near one.

A concerning path has:

- targets outside wage support;
- a sharp KL increase near full closure;
- effective-area-count ratios approaching zero;
- very large weight adjustments;
- one area receiving most of the calibrated weight;
- low observed prior mass.

Example

Gap closure	KL divergence	Effective-count ratio	Maximum adjustment	Maximum area weight
0.00	0.000	1.00	1.0	0.14
0.25	0.004	0.99	1.2	0.15
0.50	0.018	0.95	1.6	0.17
0.75	0.050	0.86	2.4	0.22
1.00	0.110	0.70	4.1	0.30

Interpretation: most of the FLS–OEWS discrepancy can be closed with modest changes to the BEA prior. Exact matching requires more concentration, but no individual area dominates the result.

The fact that the calibrated wage reaches its target is mechanical. The informative diagnostics are the KL divergence, effective-count loss, maximum adjustments, calibrated concentration, and OEWS coverage required to reach that target.